

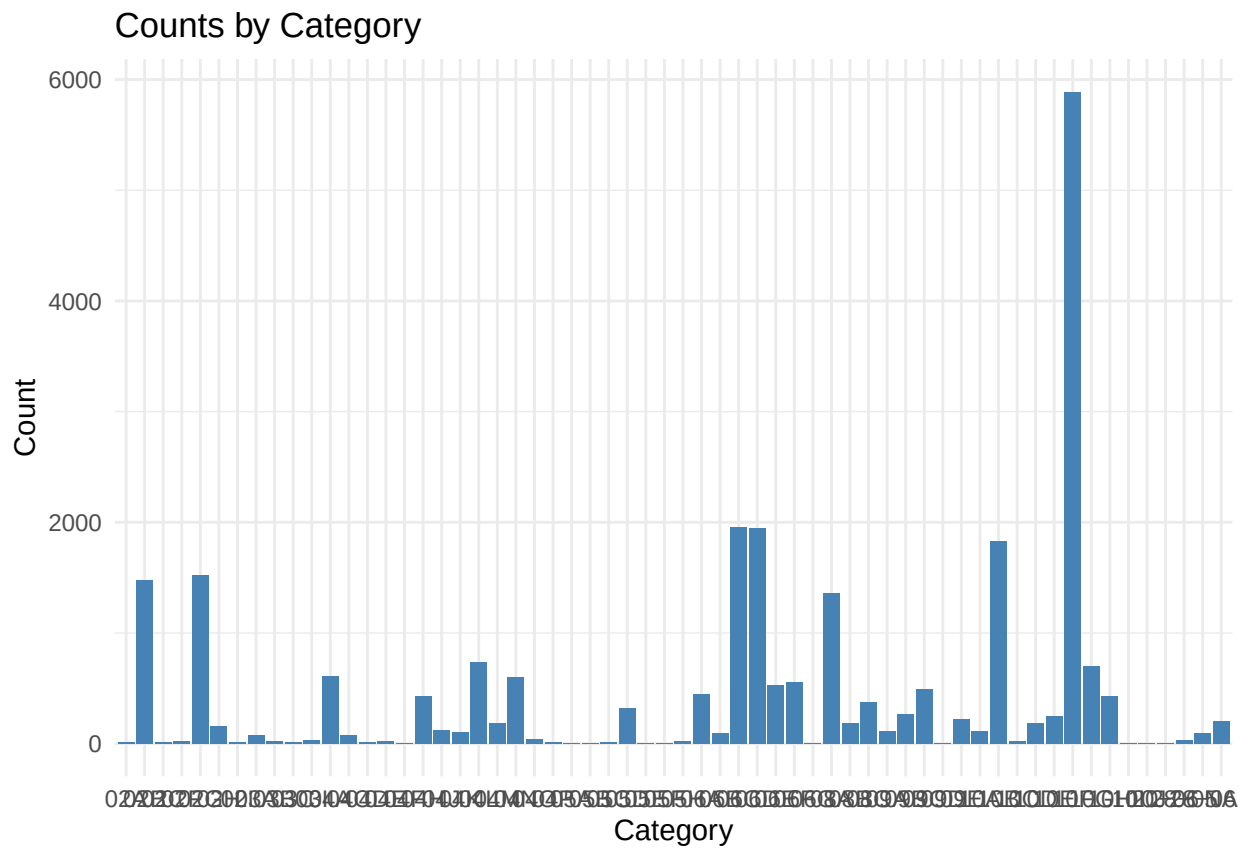
HDL Visualizations

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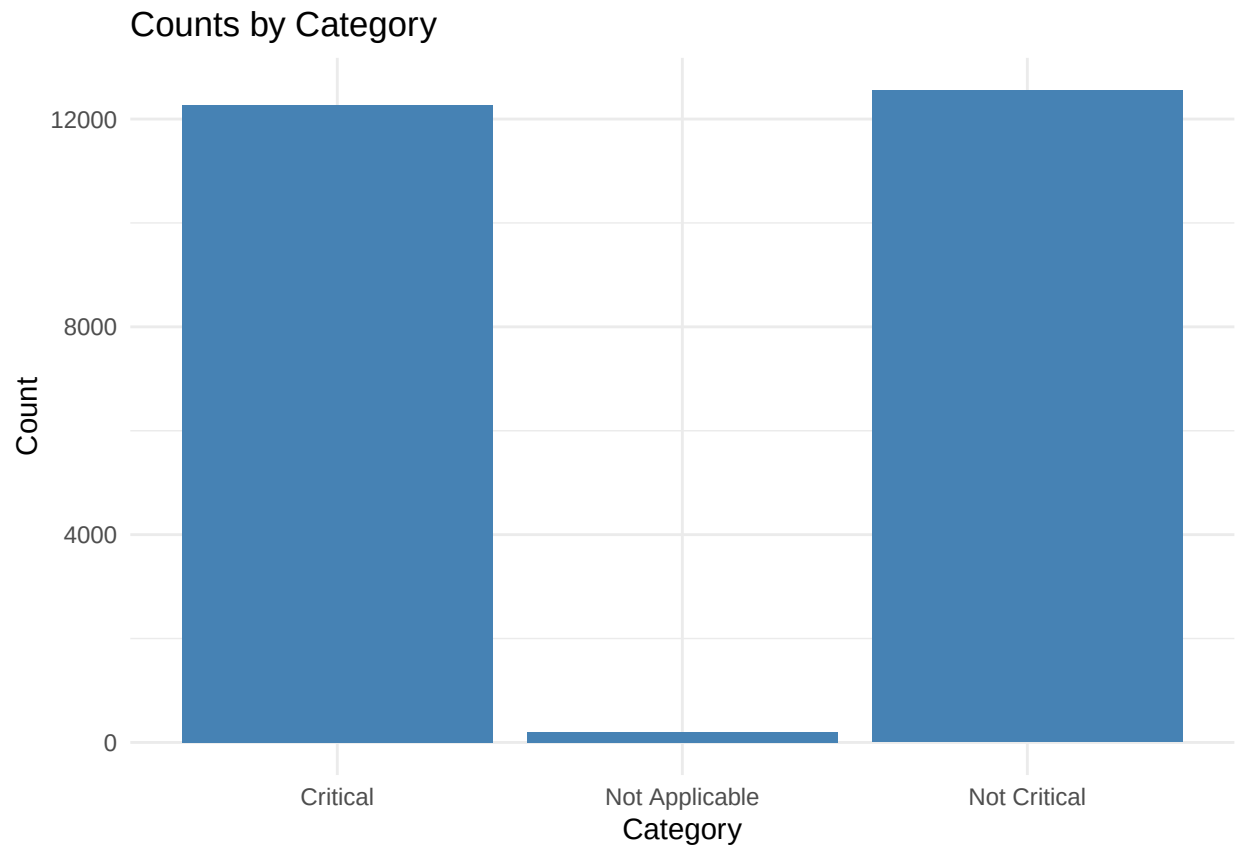
2026-07-11

Plots

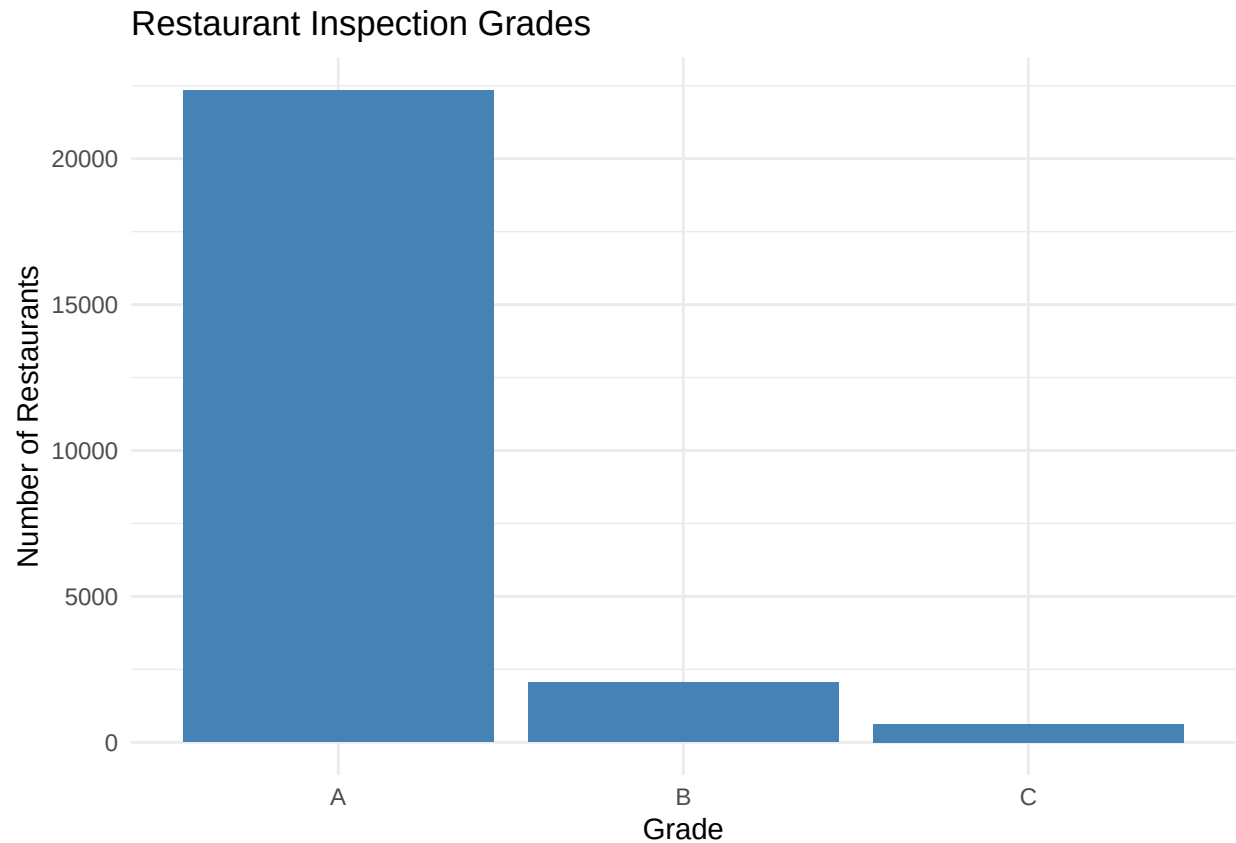
```
#violation code plot  
ggplot(restaurant_data, aes(x = violation_code)) +  
  geom_bar(fill = "steelblue") +  
  labs(  
    title = "Counts by Category",  
    x = "Category",  
    y = "Count"  
  ) +  
  theme_minimal()
```



```
#critical/not critical plot
ggplot(restaurant_data, aes(x = critical_flag)) +
  geom_bar(fill = "steelblue") +
  labs(
    title = "Counts by Category",
    x = "Category",
    y = "Count"
  ) +
  theme_minimal()
```

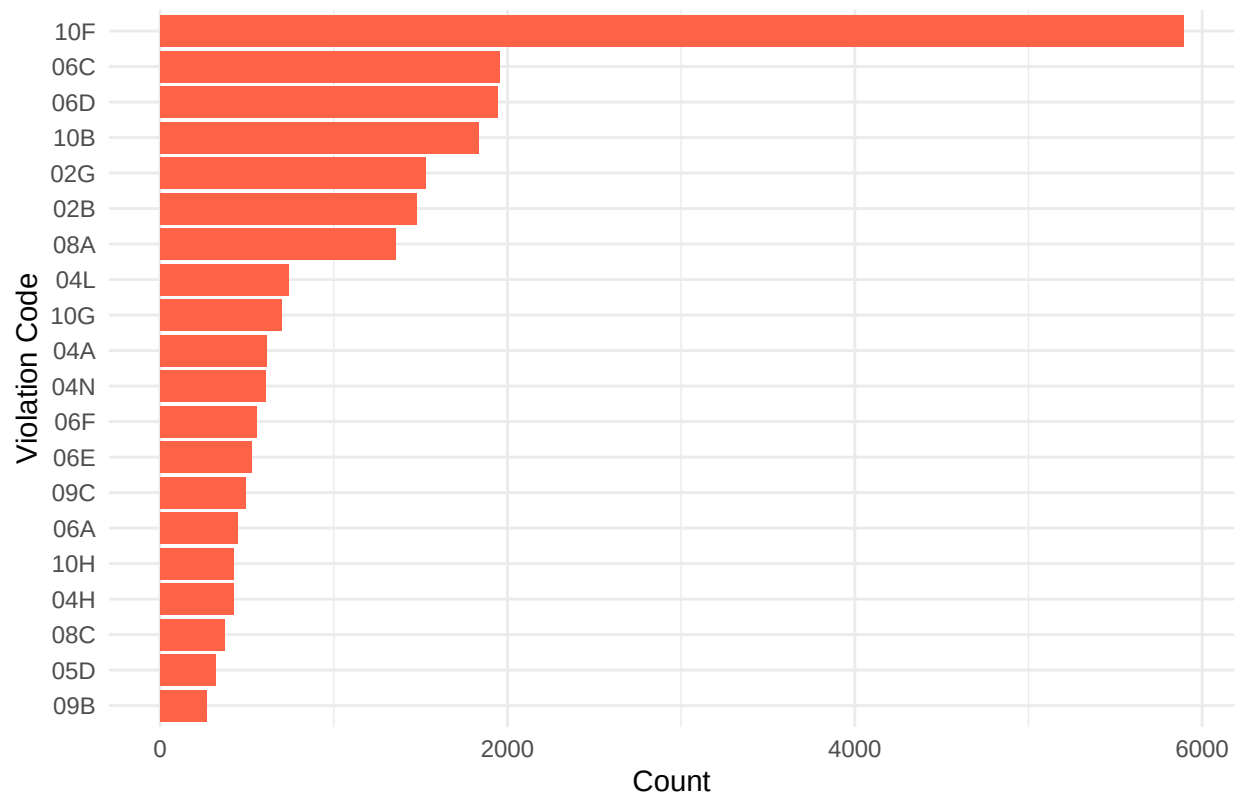


```
#restaurant grade plot
ggplot(restaurant_data, aes(x = grade)) +
  geom_bar(fill = "steelblue") +
  labs(
    title = "Restaurant Inspection Grades",
    x = "Grade",
    y = "Number of Restaurants"
  ) +
  theme_minimal()
```



```
#most common violation codes
restaurant_data %>%
  count(violation_code, sort = TRUE) %>%
  slice_head(n = 20) %>%
  ggplot(aes(x = reorder(violation_code, n), y = n)) +
  geom_col(fill = "tomato") +
  coord_flip() +
  labs(
    title = "Top 20 Most Common Violation Codes",
    x = "Violation Code",
    y = "Count"
  ) +
  theme_minimal()
```

Top 20 Most Common Violation Codes



```
#most common violation descriptions
restaurant_data %>%
  count(violation_description, sort = TRUE) %>%
  slice_head(n = 15) %>%
  ggplot(aes(x = reorder(violation_description, n), y = n)) +
  geom_col(fill = "darkorange") +
  coord_flip() +
  labs(
    title = "Top 15 Violation Descriptions",
    x = "Violation Description",
    y = "Count"
  ) +
  theme_minimal()
```

urface or equipment made of unacceptable material, not kept clean, or not properly sealed, raised, spaced or movable
Food contact surface not properly washed, rinsed and sanitized after use
nation during storage, preparation, transportation, display, service or from customer's refillable, reusable container. Co
n device not provided where required; equipment or floor not properly drained; sewage disposal system in disrepair or
r processed fish held above 38 °F; intact raw eggs held above 45 °F; or reduced oxygen packaged (ROP) TCS foods l

Establishment is n

Dishwashing and ware washing: Cleaning a

Food Prot

ciated with (FRSA) flies or other nuisance pests in establishment's food and/or non-food areas. FRSA flies include hou

W

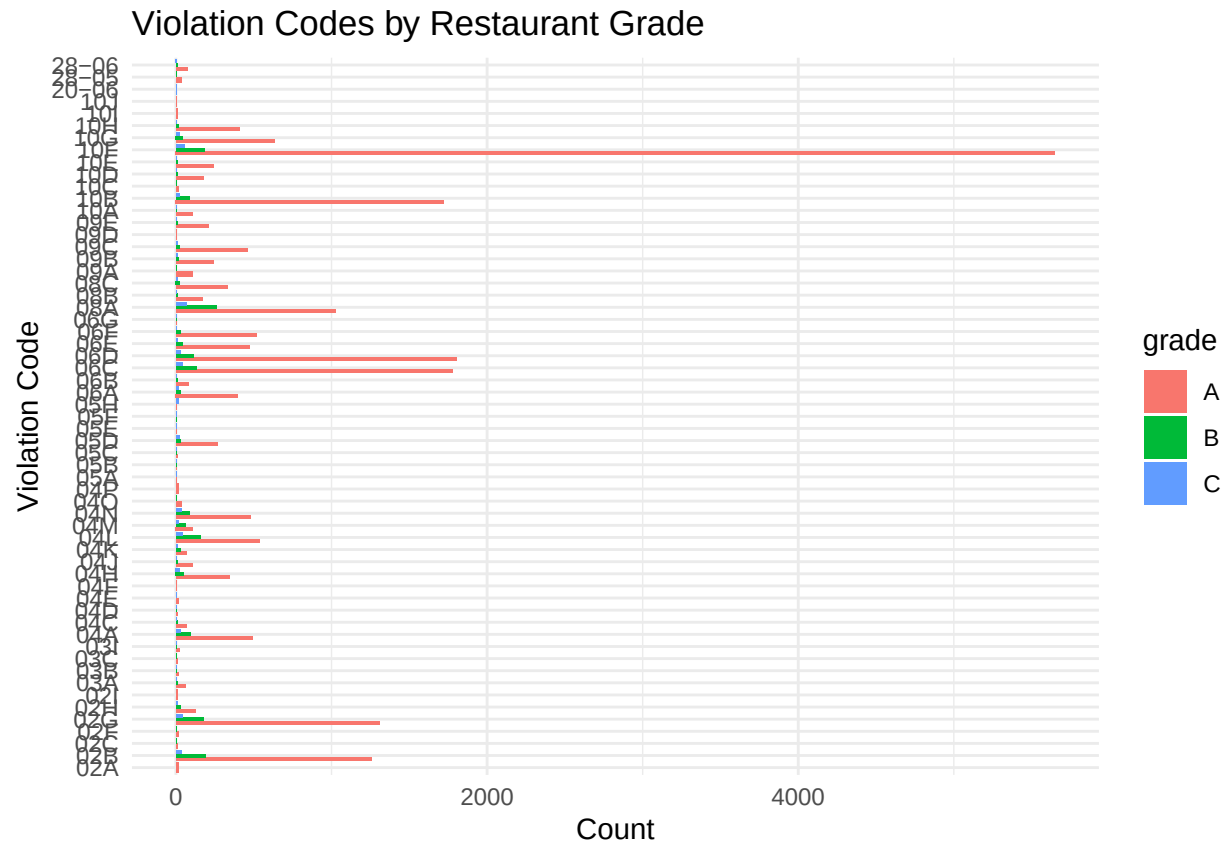
Sanitized equipment

Design, construction, materials used or maintenance of food cor

. Outer garment soiled with possible contaminant. Effective hair restraint not worn where required. Jewelry worn on ha

```
#violation codes & descriptions
View(
  restaurant_data %>%
    distinct(violation_code, violation_description) %>%
    arrange(violation_code)
)

#code and grade
restaurant_data %>%
  filter(!is.na(grade), !is.na(violation_code)) %>%
  count(grade, violation_code) %>%
  ggplot(aes(x = violation_code, y = n, fill = grade)) +
  geom_col(position = "dodge") +
  coord_flip() +
  labs(
    title = "Violation Codes by Restaurant Grade",
    x = "Violation Code",
    y = "Count"
  ) +
  theme_minimal()
```



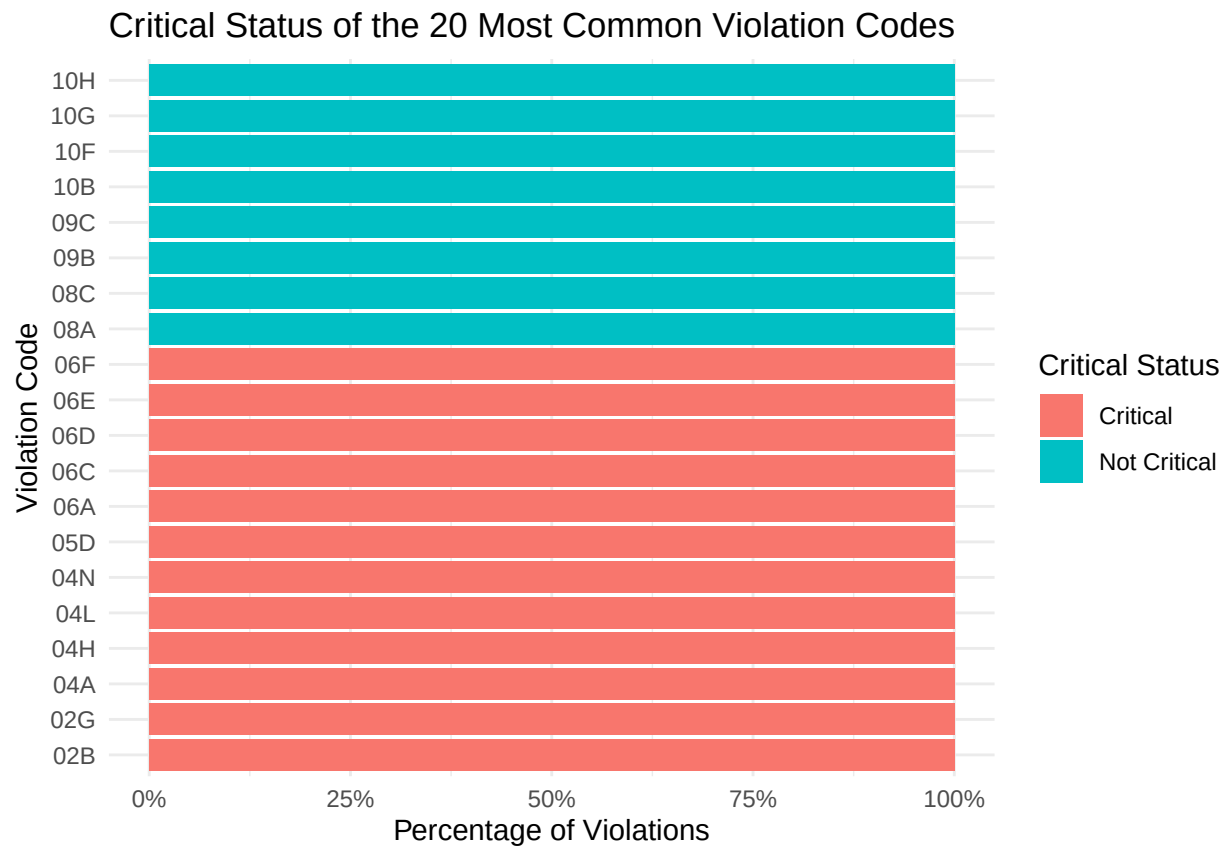
```
#grade vs critical
restaurant_data %>%
  filter(
    !is.na(grade),
    !is.na(critical_flag),
    grade %in% c("A", "B", "C")
  ) %>%
  ggplot(aes(
    x = grade,
    fill = critical_flag
  )) +
  geom_bar(position = "fill") +
  scale_y_continuous(labels = percent) +
  labs(
    title = "Critical Violations by Restaurant Grade",
    x = "Restaurant Grade",
    y = "Percentage of Violations",
    fill = "Critical Status"
  ) +
  theme_minimal()
```



```
# Find the 20 most common violation codes
top_codes <- restaurant_data %>%
  filter(!is.na(violation_code)) %>%
  count(violation_code, sort = TRUE) %>%
  slice_head(n = 20) %>%
  pull(violation_code)

# Graph critical status by violation code
restaurant_data %>%
  filter(
    violation_code %in% top_codes,
    !is.na(critical_flag)
  ) %>%
  ggplot(aes(
    x = violation_code,
    fill = critical_flag
  )) +
  geom_bar(position = "fill") +
  scale_y_continuous(
    labels = function(x) paste0(round(x * 100), "%")
  ) +
  coord_flip() +
  labs(
    title = "Critical Status of the 20 Most Common Violation Codes",
    x = "Violation Code",
    y = "Percentage of Violations",
  )
```

```
fill = "Critical Status"
) +
theme_minimal()
```



Summaries

```
#grade count & percentages
restaurant_data %>%
  count(grade) %>%
  mutate(percent = round(n / sum(n) * 100, 2))
```

```
## # A tibble: 3 x 3
##   grade     n percent
##   <chr> <int>   <dbl>
## 1 A      22343    89.3
## 2 B       2047     8.18
## 3 C        636     2.54
```

```
#violation code count
restaurant_data %>%
  count(violation_code, sort = TRUE)
```



```
## # A tibble: 60 x 2
##   violation_code      n
##   <chr>          <int>
## 1 10F            5891
## 2 06C            1952
## 3 06D            1944
## 4 10B            1834
## 5 02G            1527
## 6 02B            1478
## 7 08A            1356
## 8 04L             738
## 9 10G             702
## 10 04A            614
## # i 50 more rows
```

#10 most common violations

```
restaurant_data %>%
  count(violation_code, violation_description, sort = TRUE) %>%
  slice_head(n = 10)
```

```
## # A tibble: 10 x 3
##   violation_code violation_description      n
##   <chr>          <chr>          <int>
## 1 10F            Non-food contact surface or equipment made of unaccepta~ 5819
## 2 06D            Food contact surface not properly washed, rinsed and sa~ 1944
## 3 06C            Food, supplies, or equipment not protected from potenti~ 1918
## 4 10B            Anti-siphonage or back-flow prevention device not provi~ 1822
## 5 02G            Cold TCS food item held above 41 °F; smoked or processe~ 1519
## 6 02B            Hot TCS food item not held at or above 140 °F.          1468
## 7 08A            Establishment is not free of harborage or conditions co~ 1323
## 8 04L            Evidence of mice or live mice in establishment's food o~  727
## 9 10G            Dishwashing and ware washing: Cleaning and sanitizing o~  685
## 10 04A           Food Protection Certificate (FPC) not held by manager o~  597
```

#num of unique violations

```
restaurant_data %>%
  summarise(
    unique_violation_codes = n_distinct(violation_code),
    unique_descriptions = n_distinct(violation_description)
  )
```

```
## # A tibble: 1 x 2
##   unique_violation_codes unique_descriptions
##   <int>          <int>
## 1         60          101
```

#grade by violation

```
restaurant_data %>%
  group_by(grade) %>%
  summarise(
    inspections = n(),
    unique_violations = n_distinct(violation_code)
  )
```

```
## # A tibble: 3 x 3
##   grade inspections unique_violations
##   <chr>      <int>      <int>
## 1 A          22343          58
## 2 B          2047          47
## 3 C           636          46
```

```
#most common violation by grade
restaurant_data %>%
  group_by(grade, violation_code, violation_description) %>%
  summarise(count = n(), .groups = "drop") %>%
  arrange(grade, desc(count))
```

```
## # A tibble: 219 x 4
##   grade violation_code violation_description count
##   <chr> <chr>      <chr>      <int>
## 1 A     10F        Non-food contact surface or equipment made of una~ 5582
## 2 A     06D        Food contact surface not properly washed, rinsed ~ 1802
## 3 A     06C        Food, supplies, or equipment not protected from p~ 1749
## 4 A     10B        Anti-siphonage or back-flow prevention device not~ 1712
## 5 A     02G        Cold TCS food item held above 41 °F; smoked or pr~ 1300
## 6 A     02B        Hot TCS food item not held at or above 140 °F.     1247
## 7 A     08A        Establishment is not free of harborage or conditi~ 997
## 8 A     10G        Dishwashing and ware washing: Cleaning and saniti~ 619
## 9 A     04L        Evidence of mice or live mice in establishment's ~ 528
## 10 A    06F        Wiping cloths not stored clean and dry, or in a s~ 516
## # i 209 more rows
```

```
#grades w/violation code
restaurant_data %>%
  count(grade, violation_code) %>%
  arrange(grade, desc(n))
```

```
## # A tibble: 151 x 3
##   grade violation_code      n
##   <chr> <chr>      <int>
## 1 A     10F        5650
## 2 A     06D        1802
## 3 A     06C        1780
## 4 A     10B        1723
## 5 A     02G        1307
## 6 A     02B        1255
## 7 A     08A        1025
## 8 A     10G         636
## 9 A     04L         538
## 10 A    06F         520
## # i 141 more rows
```

```
#dataset summary
restaurant_data %>%
  summarise(
    Total_Records = n(),
```

```
Restaurants = n_distinct(camis),  
Grades = n_distinct(grade),  
Violation_Codes = n_distinct(violation_code),  
Inspection_Dates = n_distinct(inspection_date)  
)
```

```
## # A tibble: 1 x 5  
##   Total_Records Restaurants Grades Violation_Codes Inspection_Dates  
##       <int>         <int> <int>         <int>         <int>  
## 1       25026         25026     3             60             1192
```

Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.